TITLE: Covariant Quantum Hydrodynamic Regularization, Flavor Phase-Space Positioning, and Gauge Extensions at the Zepto-Scale in R^3 AUTHORS: David Botargues Martin (Independent Systems & Physics Researcher) REGISTRY: Zenodo DOI: 10.5281/zenodo.22035357 ABSTRACT: This technical memorandum delivers a complete mathematical closure for the energy density of the universe by unifying relativistic quantum hydrodynamics with non-abelian gauge field extensions. Utilizing the covariant Madelung-Klein-Gordon formalism, we rigorously confirm that a critical geometric radius R = sqrt(pi) in phase space acts as an orthogonal projector in Fourier space, interrupting the ultraviolet divergence of enstrophy within Sobolev spaces H^s(R^3). By enforcing the invariant boundary condition (K_0 = 1), the system derives a global stationary libration point characterized by the complete collapse of the material derivative of the velocity field to zero (Du/Dt = 0), shielding the continuum against entropic degradation. We demonstrate that both known Standard Model fermions and projected gauge fields are mapped onto discrete, deterministic radial coordinates (n) within the phase-space lattice, governed by the running behavior of their empirical Yukawa couplings under the Renormalization Group Equations (RGE). In the micro-scale regime (the Zeptoverso, 10^-21 m), the interaction between the high-energy Primeval Atom expansion and the backward-propagating tachyonic vacuum modes triggers a spontaneous symmetry breaking. This mechanism predicts a primary heavy vector boson (Z'_Zepto) located exactly at a center-of-mass energy of sqrt(s) = 14.0000 TeV, with a total decay width of W_Zprime = 2.7705 GeV and an integrated peak cross-section of 82.19 pb. The cosmic balance is counter-weighted at the macro-scale by a basal pseudo-scalar vacuum field (QCD axion) fixed at 6.54183024 ueV, which accounts for the exact relic dark matter density (Omega_c = 0.264) and drives the cosmological constant expansion (Omega_Lambda = 0.687) across a flat Euclidean geometry. Higher-order harmonic gauge excitations are analytically projected at discrete integer multiples of the primary mass scale threshold.